THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Arithmetic Progressions** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** -2	$1-3=-2$
2	**(c)** $-10,-6,-2,2,\ldots$	Constant difference 4
3	**(a)** 22	$a=-3,\ d=\frac{5}{2}$: $-3+10\times\frac{5}{2}=22$
4	**(a)** 6	$2(x+10)=2x+3x+2\Rightarrow x=6$
5	**(b)** Incorrect; the sum is n^2	$S_n=\frac{n}{2}[2+(n-1)2]=n^2$
6	**(a)** $4n+3$	$7+(n-1)4=4n+3$
7	**(a)** Student A is correct	$5+19\times4=81$
8	**(b)** 110	$5(4+18)=110$
9	**(a)** 30	$12,\ldots,99$: $\frac{99-12}{3}+1=30$
10	**(b)** 2	$a_n=S_n-S_{n-1}=2n+1\Rightarrow d=2$

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3+(n-1)5=78\Rightarrow n=16$; the 16th term	1+1
12	$a=7,\ d=\frac{7}{2}$; $n=\frac{84-7}{3.5}+1=23$; $S=\frac{23}{2}(7+84)=1046\frac{1}{2}$	1+1
13	$5+6(n-1)=301\Rightarrow n-1=\frac{296}{6}$, not a whole number ⇒ 301 is **not** a term	1+1
14	$a_{30}=10+29(-3)=10-87=-77$. The student is **correct**.	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$8,16,\ldots,120$: $S_{15}=\frac{15}{2}(8+120)=960$	1+1+1
16	$a+2d=4,\ a+8d=-8\Rightarrow d=-2,\ a=8$; $8-2(n-1)=0\Rightarrow n=5$: the 5th term	1+1+1
17	$\frac{n}{2}[48-3(n-1)]=78\Rightarrow n^2-17n+52=0\Rightarrow n=4$ or 13. Both valid: terms 5th to 13th add up to zero.	1+1+1
	OR: $2a+10d=24,\ 2a+14d=44\Rightarrow d=5,\ a=-13\Rightarrow -13,-8,-3$	1+1+1
18	$a=200,\ d=50$: $S_{30}=15[400+29\times50]=15\times1850=$ ₹27750	1+1+1
	OR: $5+(n-1)1.75=20.75\Rightarrow n-1=9\Rightarrow n=10$	1+1+1

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $5000+200(n-1)=7000\Rightarrow n=11\Rightarrow$ year 2005 (ii) $S_{11}=\frac{11}{2}(5000+7000)=$ ₹66000 (iii) $a_{20}=5000+19\times200=$ ₹8800 ⇒ claim false	2+2+1
	OR: $\frac{n}{2}[40-(n-1)]=200\Rightarrow n^2-41n+400=0\Rightarrow n=16$ or 25; $n=25$ gives negative top row, so 16 rows; top row $=20-15=5$ logs	3+2
20	$a+3d=7,\ a+8d=17\Rightarrow d=2,\ a=1$; $S_n=\frac{n}{2}[2+2(n-1)]=n^2$. $S_{10}=100=1+3+\cdots+19$ ✓	2+2+1
	OR: Lengths πr: $\pi(0.5+1+\cdots+6.5)=\pi\times\frac{13}{2}(0.5+6.5)=45.5\pi=\frac{22}{7}\times45.5=143$ cm	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME — SET B

Chapter 5: **Arithmetic Progressions** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{1}{3}, \frac{4}{3}$	$d = \frac{5}{3} - \frac{1}{3} = \frac{4}{3}$
2	**(a)** $35\sqrt{2}$	$\sqrt{2} + 17 \times 2\sqrt{2} = 35\sqrt{2}$
3	**(a)** $\sqrt{50}$	Terms are $k\sqrt{2}$: next is $5\sqrt{2} = \sqrt{50}$
4	**(a)** 3	$2(4k - 6) = 4k \Rightarrow k = 3$
5	**(b)** Incorrect — it can be positive, negative or zero	e.g. $10, 7, 4, \ldots$ has $d = -3$
6	**(a)** 34	$\frac{205-7}{6} + 1 = 34$
7	**(a)** Student A is correct	$\frac{20 \times 21}{2} = 210$
8	**(b)** 40	$49 - 3 \times 3 = 40$
9	**(b)** 4	Coefficient of n
10	**(b)** $n(n+1)$	$\frac{n}{2}(2 + 2n) = n(n+1)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5d = 35 \Rightarrow d = 7$, $a = 38 - 70 = -32$; $a_{31} = -32 + 30 \times 7 = 178$	1+1
12	$1, 3, \ldots, 49$: 25 terms; $S = \frac{25}{2}(1 + 49) = 625$	1+1
13	$121 - 4(n-1) < 0 \Rightarrow n > 31.25$ ⇒ 32nd term ($a_{32} = -3$)	1+1
14	$11 - 3(n-1) = -150 \Rightarrow n - 1 = \frac{161}{3}$, not a whole number ⇒ claim is **false**	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$14, 21, \ldots, 98$: $n = 13$; $S = \frac{13}{2}(14 + 98) = 728$	1+1+1
16	$a = S_1 = 8$, $S_2 = 22 \Rightarrow a_2 = 14$, $d = 6$; $a_n = 8 + 6(n-1) = 6n + 2$	1+1+1
17	$d = -2$, $n = \frac{10-34}{-2} + 1 = 13$; $S = \frac{13}{2}(34 + 10) = 286$	1+1+1
	OR: $5 + 3(n-1) = 50 \Rightarrow n = 16$; $S_{16} = 8(5 + 50) = 440$	1½+1½
18	$\frac{7}{2}[2a - 120] = 700 \Rightarrow a = 160$: ₹160, 140, 120, 100, 80, 60, 40	1+1+1
	OR: $3(1 + 2 + \cdots + 12) = 3 \times 78 = 234$ trees	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 10, 16, 22 m (there and back) (ii) $2(5 + 8 + \cdots + 32) = 2 \times \frac{10}{2}(5 + 32) = 370$ m (iii) 185 m counts only one way; return trips were ignored	1+3+1
	OR: $d = 4$, $a = 10$; $S_{51} = \frac{51}{2}[20 + 200] = 5610$ (3). $a_{26} = 10 + 100 = 110$; $51 \times 110 = 5610$ ✓ (middle-term property) (2)	3+2
20	$2a + 5d = 12$, $2a + 15d = 32 \Rightarrow d = 2$, $a = 1$; $S_{10} = 5(2 + 18) = 100$; $a_{20} = 1 + 38 = 39$	2+2+1
	OR: $7d = 7 \Rightarrow d = 1$; $S_{20} = 10(6 + 19) = 250$; $3 + (n-1) = 50 \Rightarrow n = 48$ ⇒ yes, 48th term	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.